

Classifying building age in a city with almost no local labels



Madrid teaches the model. **Amsterdam** gives it as few as 5 labelled pixels per class.

4

building-age classes

60

ordered spectral features

macro-F1

equal weight, every class

WHAT MAKES IT HARD

Domain shift between cities, mixed-pixel class noise, and support sets as small as 5 examples per class.

WHAT SUCCESS MEANS

Strong, equally-weighted macro-F1 across every budget — reproducible, and honest about where it does not generalize.

One pipeline, five stages

From two cities of spectral data to a single calibrated prediction — the whole workflow at a glance.



The name: “Coordinate-RF” refers to stages 3–4 — a Random Forest trained on the small Amsterdam support set, with two standardized lattice coordinates appended so it can use local geography, then blended with the source-city prior.

Two cities, one clean protocol

DATASET

- Source: Madrid — fully labelled, 60 spectral features per pixel.
- Target: Amsterdam — small, class-balanced support set.
- Local model adds 2 standardized lattice coordinates (x, y).
- Support budgets tested: 5, 25, 50, 100, 200 labelled pixels per class.

WHY THE PROTOCOL IS VALID — NO LEAKAGE

- Pseudo-labels used for source alignment come only from the source model — the query set's true labels are never touched.
- The final query population excludes both the current support set and a separate 800-label development bank.
- Support and query pixels stay strictly separated through every stage of inference.
- Every run is matched: 200 paired episodes per budget, same pixels seen by every method compared.

Coordinate-RF: three design choices

Each choice targets the same constraint — very few Amsterdam labels.

1

Why Random Forest

A balanced, 200-tree forest fits reliably on support sets as small as 5 labels per class — no gradient tuning, minimal overfitting risk.

2

Why coordinates

Two standardized lattice coordinates are appended to the local model's inputs, letting splits exploit the fact that nearby pixels tend to share a building age.

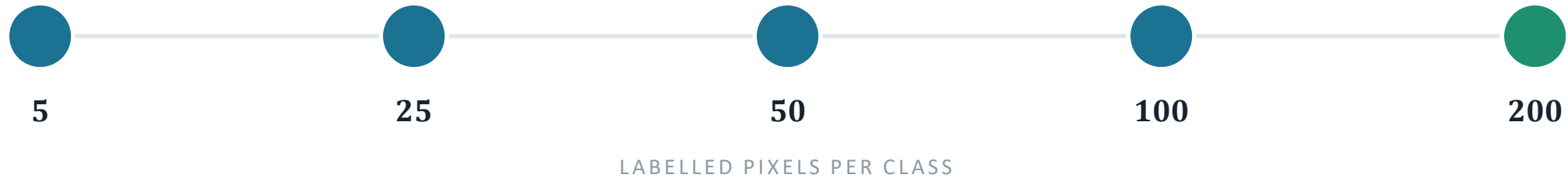
3

Why blending helps

A source-trained probability prior fills the gap when local support is tiny. Its weight, $20/(20+4b)$, shrinks automatically as budget b grows.

Then: the blended probabilities are smoothed over 9 spatial neighbours with Gaussian weights, stabilizing predictions before evaluation.

Five budgets, evaluated the same way every time



200 paired episodes

Every budget is evaluated across 200 matched episodes, with population SD reported alongside each score.

Macro-F1, equal weight

The four age classes are weighted equally — a rare class can't be ignored to inflate the score.

Same pixels, every method

Coordinate-RF, ASTRA, EXP-F and EXP-010 are all compared on identical, paired support/query splits.

0.749299 macro-F1 — the best score at every budget

LOCKED RESULT · 200 SHOTS

0.7493

macro-F1 ±0.005716 population SD

Matched gains at 200 shots, all 200 paired episodes:

vs ASTRA

+0.0163

200/200 episodes won

vs EXP-F

+0.0917

vs EXP-010

+0.1265

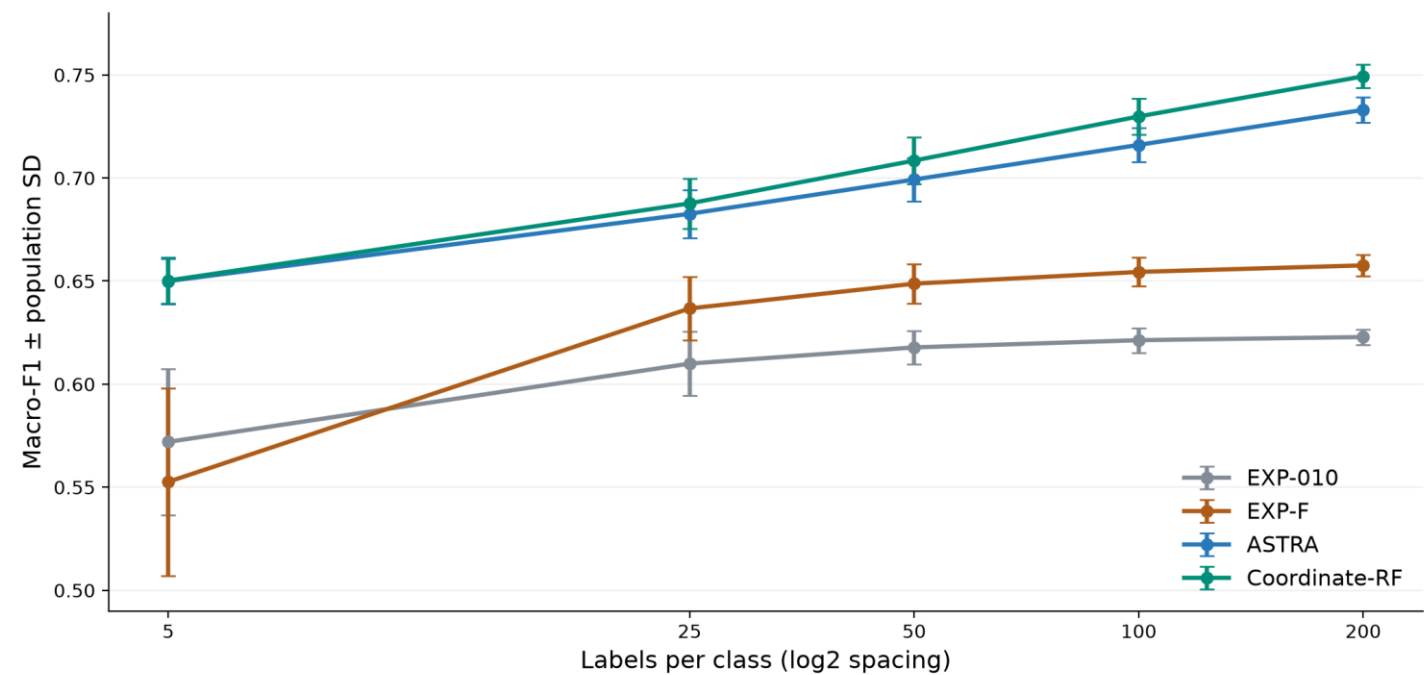
Budget / class	Macro-F1	Population SD
5	0.650253	±0.011246
25	0.687664	±0.012145
50	0.708436	±0.011277
100	0.729839	±0.008802
200	0.749299	±0.005716

Why it matters: Coordinate-RF isn't just the best score at 200 shots — it's the top locked result at every budget tested, including the hardest, 5-shot regime.

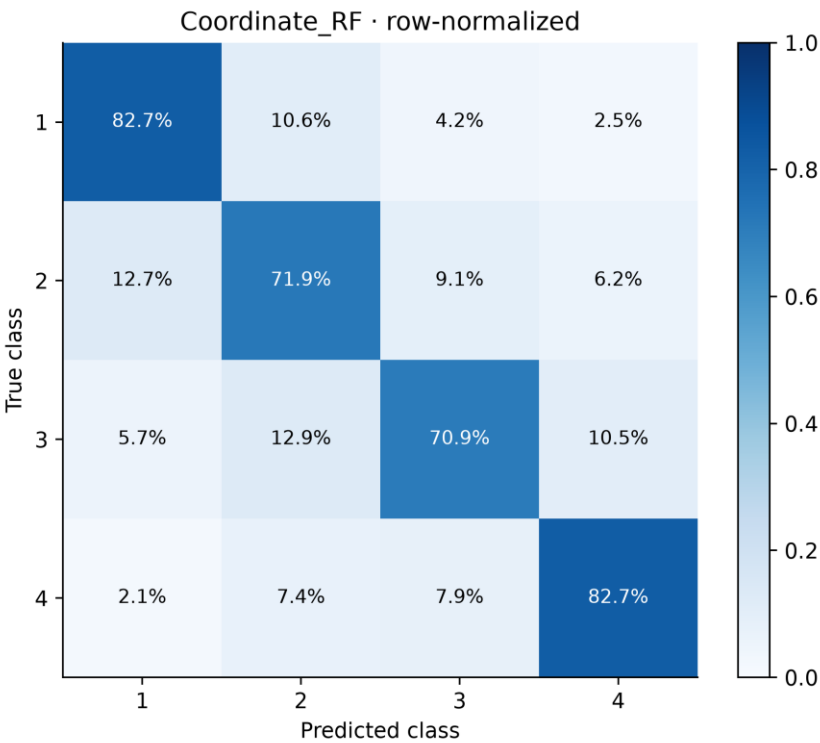
95% BOOTSTRAP CI · 200 SHOTS

[0.7485, 0.7501] macro-F1, resampled from the same 200 paired episodes — the headline number is not a single lucky draw.

Consistent gains as labels grow

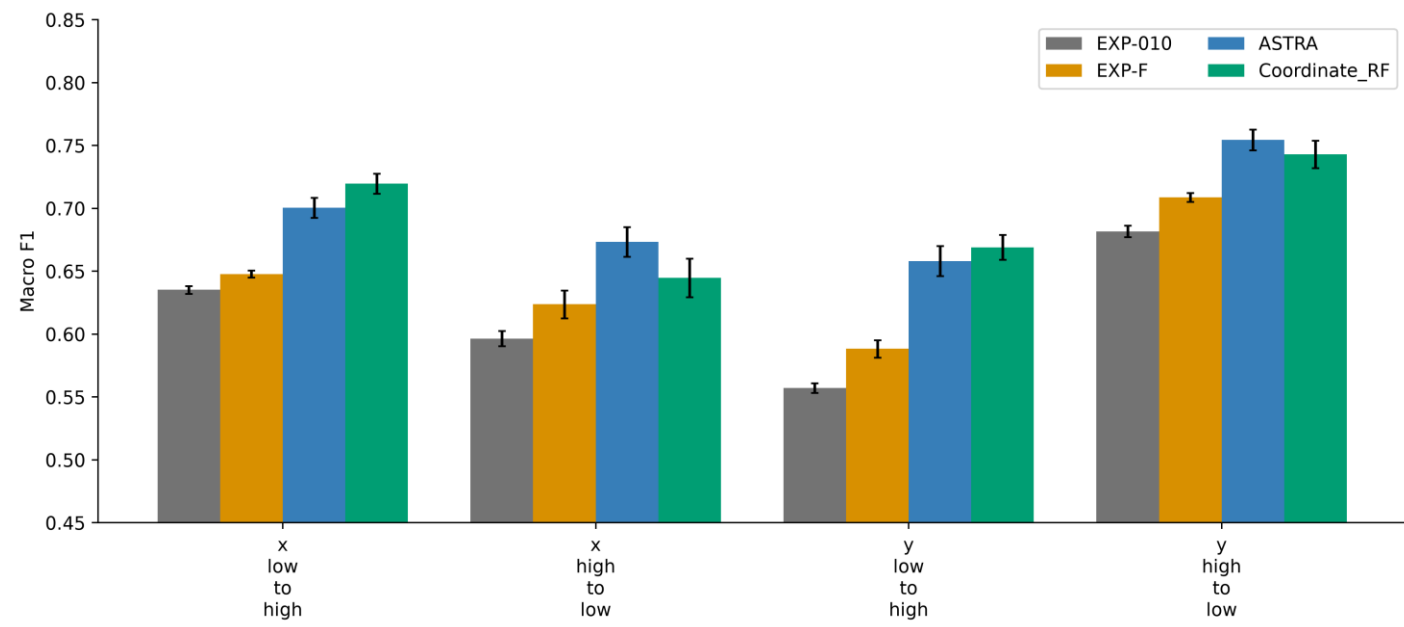


Learning curve: macro-F1 rises smoothly from 5 to 200 shots, with Coordinate-RF (green) ahead of every baseline at every budget.



What the model learned: 71–83% correct per class, with confusion concentrated between adjacent age classes — the pattern we'd expect.

Where the method is honest about its limits



Spatial stress test: held-out on the opposite half of the city. Mean 0.6942 vs ASTRA's 0.6967; worst direction 0.6448 vs ASTRA's 0.6582 — coordinates help interpolation between nearby labels, not extrapolation far beyond them.

Transductive, not inductive

Full-pool, coordinate-aware inference bound to the supplied Amsterdam pool — not an unseen-city claim.

Pool-bound artifact

Excludes current support and an 800-label development bank; a new pool needs re-fitting on organizer data.

Future work

Extend beyond interpolation so the model holds up further from its labelled coordinates.

Also disclosed: the inherited ASTRA baseline recipe carries historical audit exposure; no independent organizer-held-out validation is claimed here, and protocol eligibility must still be confirmed against organizer rules.

Why this is useful beyond one leaderboard score



Scales gracefully with labels

Macro-F1 climbs smoothly from 5 to 200 shots — useful whenever labelling a new city is slow or expensive, since even a handful of examples is workable.



Fully reproducible

Frozen configurations, hashes, label-free prediction replay and saved confusion matrices mean the result can be independently checked, not just trusted.



A transferable blueprint

Source prior + local coordinate-aware fit is a pattern that extends to other cross-city, low-label spectral classification problems within the same random-pixel, full-pool setting.

What to remember about Coordinate-RF

0.749299

Locked macro-F1 at 200 shots — the best result at every budget tested, 5 through 200.

200 / 200

Matched episodes where Coordinate-RF beat the strongest baseline, ASTRA, at 200 shots.

Zero leakage

Query truth never touches training, pseudo-labelling, or model selection — fully reproducible.

Honest scope

Transductive and pool-bound by design; coordinates aid interpolation, not extrapolation.